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EMPIRICAL ANALYSIS: SVR WITH DIVERSE KERNEL MODELS FOR FORECASTING AMAZON STOCK PRICE

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ABSTRACT

Current stock price prediction methods often struggle with non-linear market behavior. This paper addresses this limitation by proposing a Support Vector Regression (SVR) approach with multiple optimized kernel functions for improved Amazon stock price prediction from 1997 to 2023. This research explores the effectiveness of four kernels: wavelet, Ensemble Hyperbolic Tangents with Bayesian Optimization (EHT-BO), Combined Polynomial Radial Basis Function (RBF) with Grid Search (CPR-GS), and Hybrid ANOVA Polynomial RBF with Grid Search (HAPR-GS). The models are evaluated using various regression metrics. Among them, the CPR-GS kernel SVR model achieves the best performance with a Mean Absolute Error (MAE) of 0.159, indicating its potential for accurate stock price prediction.

KEYWORDS: Amazon stock price, SVR, wavelet, EHT-BO, CPR-GS, and HAPR-GS kernels.

1. INTRODUCTION

The market capitalization of Amazon, as of December 2023, exceeded $1.59 trillion, solidifying its position as one of the world's largest e-commerce and technology companies. Amazon's stock price reflects its performance, growth, and innovation, while also being influenced by market expectations and sentiments. Predicting Amazon's stock price is crucial and challenging for investors, traders, and analysts, as they strive to make informed decisions and optimize their strategies. However, this task is complicated due to the impact of various variables, including market trends, company performance, news events, and investor sentiment [1]. Traditional statistical and econometric methods, such as linear regression and autoregressive integrated moving averages (ARIMA), may not fully capture the intricate dynamics and patterns of stock price movements [5].

This paper introduces a novel approach for predicting Amazon's stock price using support vector regression (SVR) with multiple kernel methods. The SVR algorithm identifies a hyperplane that best fits the data points while allowing for some deviations within a predetermined margin [2]. By leveraging kernel functions, SVR can effectively handle nonlinear and high-dimensional data, mapping it into a higher-dimensional space and measuring the similarity between input vectors. The choice of kernel function and its parameters significantly impact model performance and efficiency. Our primary objective is to forecast Amazon's stock price [4]. To achieve this, Amazon dataset utilized spanning from 1997 to 2023.

The dataset has applied to four different kernel functions of SVR, each fine-tuned with optimization techniques for tuning hyperparameters: wavelet, ensemble hyperbolic tangent with Bayesian optimization (ETH-BO), combined polynomial Radial basis function (RBF) with grid search (CPR-GS), and hybrid ANOVA polynomial RBF with grid search (HAPR-GS). Later, evaluate these models based on five key metrics: mean absolute error (MAE), mean square error (MSE), root mean square error (RMSE), mean absolute percentage error (MAPE), and the coefficient of determination (R^2 score). The results obtained from SVR models employing different kernel functions are compared to perform the analysis.

The rest of the paper is structured as follows: Section 2 presents related research on stock price prediction using SVR kernel functions. The methodology for the paper is described in Section 3, which includes data

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collection and preprocessing, model formulation and parameter tuning, and evaluation metrics. The results of the experiment are presented in Section 4 and discussed. Finally, Section 5 concludes the paper by suggesting some areas for future research.

2. RELATED WORK

The following are some of the studies that used SVR and a variety of kernel functions to make stock price predictions:

According to Dash R. K et al. (2021) [2], SVR can predict the Tata Steel stock price using linear, RBF, and polynomial kernels. They evaluated the models using MAE, RMSE, MAPE, and R^2 scores metrics. RBF kernel outperformed the polynomial and linear kernels in terms of all metrics. SVR models outperformed ARIMA models for time series forecasting, which is a traditional statistical method.

M V Kamal and D Vasumathi (2019) [4] develop a framework for predicting stock prices using SVR and Monte Carlo methods. The reason they use SVR with RBF is that it has a high capacity, a simpler design, and can be used as a kernel function. To evaluate the efficiency of the proposed method, four companies' stock prices are considered: Microsoft, Facebook, Amazon, and Google. SVR-Monte Carlo demonstrated efficient error value calculation in experimental results. Using SVR-Monte Carlo, the MSE on Google stock was 0.2162 and the MAE was 0.0164.

A study conducted by Ouahilal M et al. (2017) [7] used daily data from 2010 to 2015 to predict Google's stock price using SVR with linear, RBF, and polynomial kernels. Evaluation metrics included MAE, RMSE, and R^2 score. The researchers found that the RBF kernel was the most effective overall, followed by the polynomial kernel and the linear kernel. In addition, they found that SVR models performed better than multiple linear regression (MLR), a well-known and commonly used regression model.

An international trade prediction was done by Subba reddy et al. (2023) [8] based on the imports and exports of various commodities available in the countries. A wavelet-based SVR algorithm has been proposed for forecasting trade value. The authors attempted to predict imports and exports of various commodities of the countries using proposed wavelet and RBF kernels of SVR, the wavelet kernel exhibits better predictions for split ratios of 90:10, 80:20, 70:30, 60:40, and 50:50 during training and testing.

The ensemble hyperbolic tangent kernel-SVR model (HTK-SVR-BO) was developed by Subba reddy et al. (2024) [9] using Tanh and inverse Tanh kernels with Bayesian optimization (BO). A second ensemble SVR model (LPR-SVR-BO) uses well-known kernel types, such as linear, polynomial, and RBF kernels. Microsoft Corporation's stock prices are analyzed using the proposed models. As compared to LPR-SVR-BO, HTK-SVR-BO performs better according to the regression metrics MAE, MSE, RMSE, R^2 score, and MAPE, with results of 0.27424, 0.13392, 0.36595, 0.99997, and 5.2331.

Subba reddy et al. (2024) [10] propose an effective method for predicting the stock closing prices of three popular cryptocurrencies: Bitcoin, Ethereum, and Polkadot. Author compares RBF kernel and the hybrid kernels designed based on cryptocurrency data characteristics. The grid search technique can also be used to tune SVR hyperparameters. An evaluation of the author's approach, the proposed hybrid kernels with SVR got a MAPE value of 0.07772 and an R^2 score of 0.9999.

The RBF kernel has been shown to be the most effective and efficient kernel function for predicting stock prices with SVR. In addition, some studies have shown that some kernel methods are more effective than RBF. This study compares those effective kernel methods and finds which one is more suitable for predicting stock prices through SVR.

3. METHODOLOGY

Our proposed work involves collecting historical daily stock prices of Amazon [11] from Yahoo Finance website spanning from May 16, 1997 – December 30, 2023. This dataset includes essential attributes such as dates, opening prices, high prices, low prices, close prices, adjusted close prices, and volume. The close price serves as the target attribute for the proposed machine learning (ML) models. The collected dataset comprises 6,700 records. After examining the correlation matrix, found that all attributes except volume attribute, exhibit strong correlations. Subsequently, eliminated the least correlated attribute from the analysis. With this refined dataset, the proposed ML models were trained. Specifically, four kinds of proposed SVR kernel models are analyzing the Amazon dataset. Before feeding the dataset into these kernel models, the dataset is split it into train and test datasets using various ratios (e.g., 90:10, 80:20, 70:30, 60:40, and 50:50).

Figure 1 provides an overall block diagram of our proposed approach, while Figure 2 showcases schematic

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representations of various kernel models for SVR.

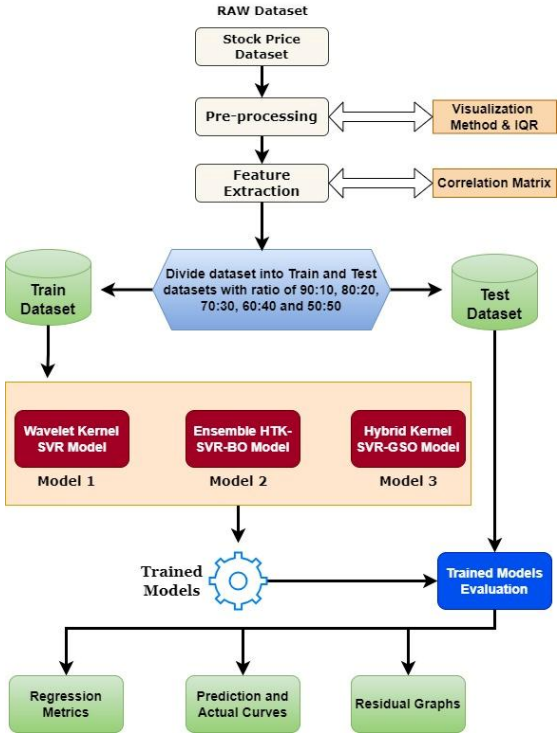


Figure 1. *Amazon stock price prediction based on diverse kernel SVR models*

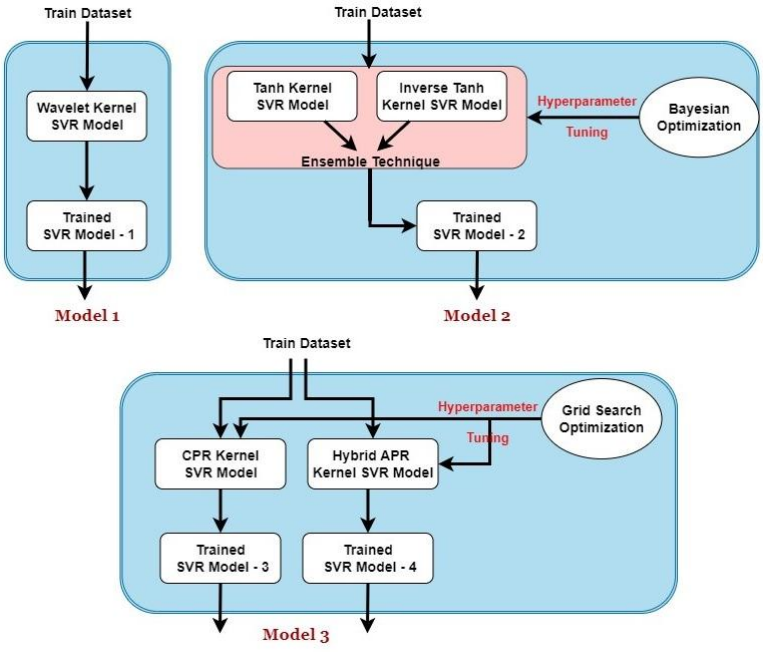


Figure 2. *Schematic representation of Diverse kernel - SVR models*

In first model utilizes a wavelet kernel instead of linear, polynomial, or RBF kernels [8]. The second model incorporates hyperbolic tangent functions such as tanh() and tanh-1(). Each of these functions has its own unique set of features. To integrate these functions, an ensemble technique (bagging) is applied. An ensemble hyperbolic tangent function is then considered as a kernel function for SVR. Later, Bayesian optimization (BO) approach was used to tune the hyperparameters of the SVR kernels [9].

The third model considers two kernel functions, in that first function is polynomial and RBF kernels are

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averaged. In the second function, ANOVA, polynomial, and RBF kernels are combined based on adaptive tuning method. For tuning these kernel parameters of the SVR for stock close price prediction, grid search optimization (GSO) has been applied [10].

3.1. Support Vector Regression

In the context of regression tasks, SVR is a machine learning (ML) technique designed to predict continuous output values. Leveraging principles from Support Vector Machines (SVM), SVR aims to minimize the error between actual and predicted values by defining a hyperplane while allowing for a fixed margin of error. To capture intricate relationships and non-linear patterns, the kernel trick maps input data into a higher-dimensional space. SVR also incorporates an epsilon-insensitive loss, considers support vectors that influence the optimal hyperplane, and employs various kernel functions (such as linear, polynomial, and radial basis functions) to accommodate diverse data types. Through fine-tuning regularization parameters and epsilon, SVR strikes a balance between model complexity and accuracy. In finance, time series forecasting, and regression scenarios demanding the capture of complex patterns, SVR effectively handles nonlinearity.

It is possible to write the decision function for SVR as follows:

$$f(x_i) = \sum_{i=1}^{l}(\alpha_i^* - \alpha_i)K(x_i, x_j) + b \quad (1)$$

Where b is constant vector, α_i and α_i^* are Lagrange multipliers with kernel function K, where $K(x_i, x_j) = \Phi(x_i) \cdot \Phi(x_j)$.

3.2 Kernel Functions

In the current research, the following kernel functions are used in the SVR model:

3.2.1 Wavelet kernel [12]

According to Mercer's theorem, the wavelet kernel is valid as a kernel function. It emphasizes the importance of the mother wavelet h (x) and how it interacts with the translation and dilation parameters $\boldsymbol{c}$ and $\boldsymbol{a}$. Here $x, a, c \in R$, and if $X, Y \in R^N$, the dot-product kernels are [3]

$$K(\mathrm{X}, \mathrm{Y}) = \prod_{i=1}^{N} h\left(\frac{X-c_i}{a}\right) h\left(\frac{Y-c_i^*}{a}\right) \quad (2)$$

Thus, wavelet kernels satisfying the translation invariant kernel theorem are of major importance

$$K(\mathrm{X}, \mathrm{Y}) = \prod_{i=1}^{N} h\left(\frac{X-Y}{a}\right) \quad (3)$$

3.2.2 Hyperbolic tangent kernels [9]

The kernels are derived from the sigmoid activation function, which is commonly applied to artificial neural networks. SVR uses Tanh kernels to model complex patterns and make accurate predictions even when dealing with complex data sets.

Hyperbolic Tangent Kernel

$$K(X, Y) = \tanh\left(p_1 + \left(p_2 * (X.Y')\right)\right) \quad (4)$$

Inverse Hyperbolic Tangent Kernel

$$K(X, Y) = \tanh^{-1}(p_1 + (p_2 * (X.Y'))) \quad (5)$$

3.2.3 CPR & Hybrid APR kernels [10]

In the realm of ML, the creation of a novel kernel called CPR (combined polynomial and radial basis function (RBF) kernel) offers practitioners a versatile and powerful tool for SVR model. CPR enhances flexibility, boosting generalization performance, improving noise tolerance, and ensuring computational efficiency, all while enhancing interpretability. Additionally, another hybrid kernel known as HAPR (hierarchical ANOVA, polynomial, and RBF kernel), proves effective in both ML and statistical modeling. Notably, HAPR extends beyond CPR's kernel features, making it applicable to a wider range of data types.

Polynomial kernel

$$K_{Poly}(X, Y) = (p_2 \cdot X^T Y + p_1)^d \quad (6)$$

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RBF kernel

$$K_{RBF}(X,Y) = \exp(-p_2.\|X-Y\|^2) \quad (7)$$

ANOVA kernel

$$K_{Anova}(X,Y) = \sum_{i=1}^{n} \exp(-p_2.(X_i - Y_i)^2)^d \quad (8)$$

CPR Kernel

$$K(X,Y) = (K_{Poly} + K_{RBF}) / 2 \quad (9)$$

HAPR Kernel

$$K(X,Y) = \alpha.K_{RBF} + (1-\alpha).(K_{Anova} + K_{Poly}) \quad (10)$$

Where X and Y attributes are values of independent variables, d, p_1, p_2 and α are the kernel parameters.

3.3 Optimization & Ensemble techniques

3.3.1 Grid Search

Grid search involves defining a grid of hyperparameter values. For each combination of these hyperparameters, the models were trained and evaluated [6]. The goal is to find the optimal setting values by identifying the hyperparameter combination that yields the best performance. Grid search is straightforward to implement and can be applied to various ML models. However, it's important to note that while it offers some computational efficiency, running grid search can become expensive, particularly when dealing with a large number of hyperparameters.

3.3.2 Bayesian Optimization

Bayesian optimization tunes hyperparameters by using a probabilistic model to assess their impact on model performance [9]. This model guides the exploration of an initial set of hyperparameter combinations iteratively, selecting those that seem promising results. As each experiment is completed, the model is updated, and the process continues until a convergence criterion is met. Compared to grid search, the Bayesian approach proves more efficient, especially when dealing with a large number of hyperparameters or a high-dimensional grid. Additionally, it offers valuable insights into the relationship between hyperparameters and model performance, helping us understand how the model behaves.

3.3.3 Ensemble Technique

In machine learning, bagging, or bootstrap aggregating, improves predictive model performance by combining the predictions of multiple base models. This method utilizes a bootstrap sampling technique to create multiple subsets of the training data. These subsets, which may contain duplicates, lead to a more diverse dataset. Each subset is then used to train independent base models, such as decision trees in the popular Random Forest algorithm. Finally, an aggregation process, like a majority vote for classification or averaging for regression, are determines the final prediction.

3.4 Evaluation metrics

The correctness of a model for the original time series can be determined by evaluating errors using metrics like MAE, MSE, RMSE, MAPE, and R^2 score [9].

$$MAE = \frac{1}{M}\sum|z_i - \hat{z}_i| \quad (11)$$

$$MAPE = \frac{1}{M}\sum\frac{|z_i - \hat{z}_i|}{z_i} \quad (12)$$

$$MSE = \frac{1}{M}\sum(z_i - \hat{z}_i)^2 \quad (13)$$

$$RMSE = \sqrt{MSE} = \sqrt{\frac{1}{M}\sum(z_i - \hat{z}_i)^2} \quad (14)$$

$$R^2\ score = 1 - \frac{\sum(z_i - \hat{z}_i)^2}{\sum(z_i - \bar{z}_i)^2} \quad (15)$$

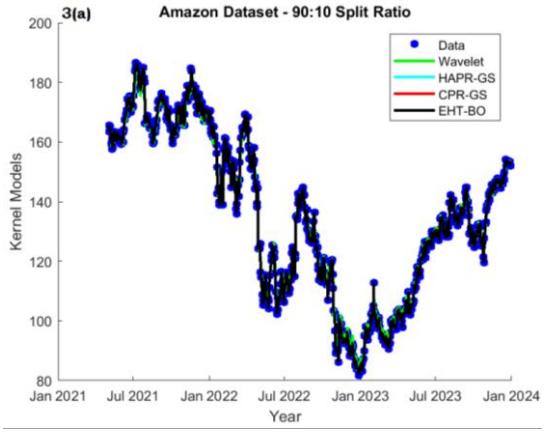
Where M is the total number of records, z_i is the actual value, $\hat{z}_i$ is the predicted value, and $\bar{z}_i$ is the mean value of z_i

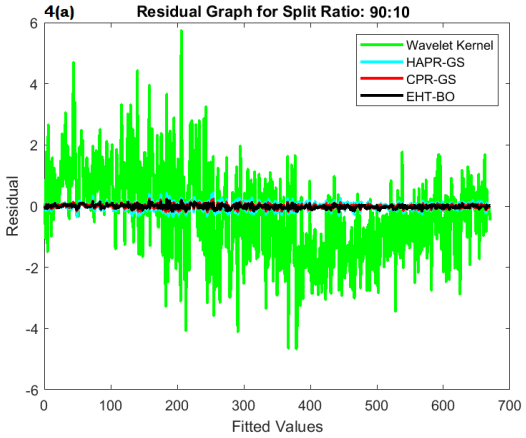
4. RESULTS AND DISCUSSION

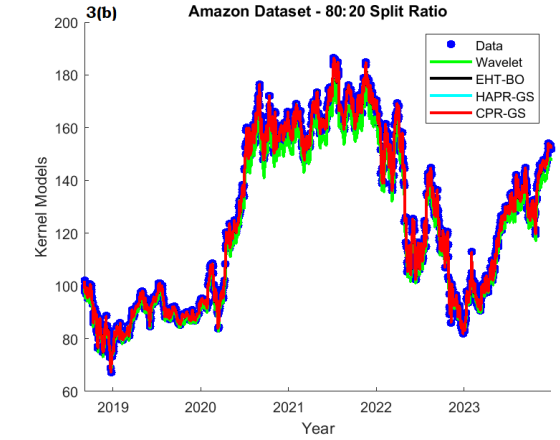
Our research utilized Amazon time series datasets for both training and testing machine learning models. Within the SVR models, a variety of kernels are explored, including wavelet kernels, EHT-BO, CPR-GS, and

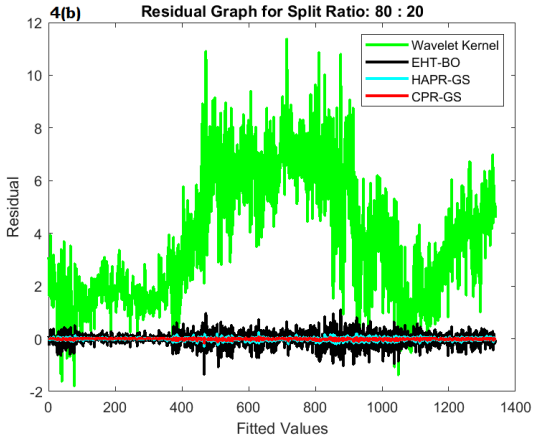
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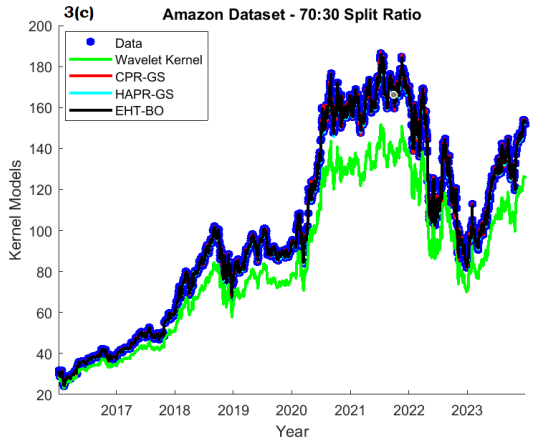
HAPR-GS, all optimized to enhance performance. The results were visualized in the form of prediction curves (PCs) and residual graphs (RCs) for diverse kernel models (as depicted in Figures 3(a)-3(e) and Figures 4(a)-(e)). These curves provide insight into how model complexity impacts performance. Specifically, the PC graph includes four kernel method curves alongside the actual curve. These curves help us to understand how model complexity and predictive accuracy interact. The RC graph highlights the discrepancies between the test dataset values and the predicted values. Residual plots serve as valuable tools in regression analysis, aiding in diagnosing potential issues with a model's fit to the data. This wide-ranging approach allows us to assess the efficacy of different SVR kernel methods and their impact on stock price prediction.

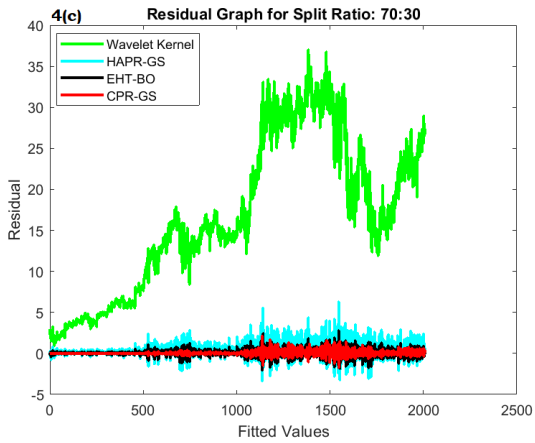




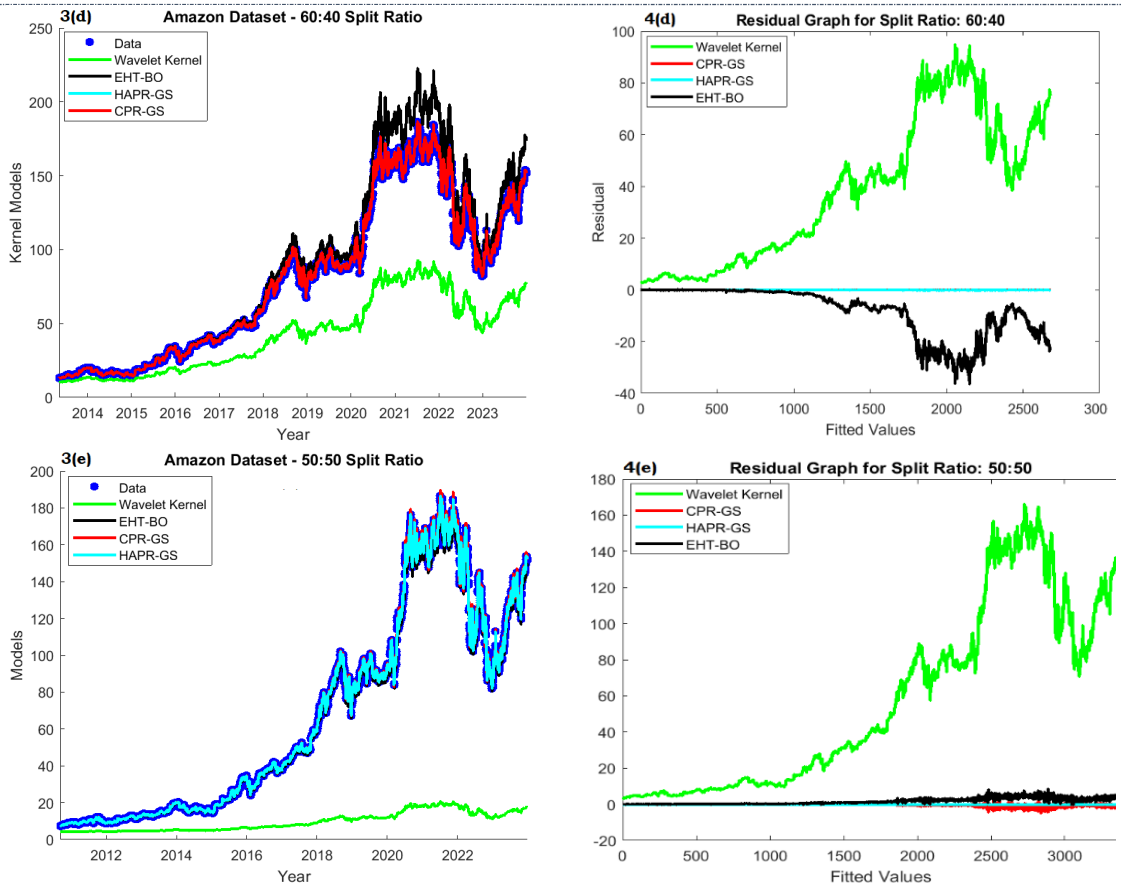








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Figures 3(a)-3(e). *Predictive cureves for various split ratios of Amazon dataset*
Figures 4(a)-4(e). *Residual cureves for various split ratios of Amazon dataset*

Our investigation into Amazon stock price prediction employed various SVR kernel methods. We evaluated their performance using prediction curves (PCs) and residual graphs (RCs). Wavelet kernel exhibited the poorest performance across all split ratios for the Amazon datasets. Its predictions deviated significantly from the actual stock prices.

For the 90:10 split, both CPR-GS and EHT-BO produced comparable PCs. However, EHT-BO displayed a clear advantage in terms of RCs, with noticeably fewer residuals. This suggests EHT-BO as the most efficient and effective choice for the Amazon dataset in this specific split ratio.

For the 80:20 and 70:30 split ratios, EHT-BO, HAPR-GS, and CPR-GS yielded similar PCs. However, RC analysis revealed a clear ranking based on residual values, from highest to lowest: wavelet, EHT-BO, HAPR-GS, and CPR-GS. Consequently, CPR-GS emerged as the more efficient choice for these split ratios.

In the 60:40 split, neither CPR-GS nor HAPR-GS demonstrated clear efficiency based on PC and RC analysis. Other kernel methods also proved ineffective in this split ratio. For the 50:50 split, both RCs and PCs showed minimal variation among the kernel methods.

Compared to HAPR-GS, EHT-BO and CPR-GS displayed greater fluctuations in prediction curves (PCs) for 2021 data, and their residuals increased significantly since record number 2000. These observations position HAPR-GS as the superior performer for this split ratio. These findings provide valuable insights for selecting the most suitable SVR kernel method for accurate stock price predictions, considering both efficiency and stability.

4.1 Comparison

Table 1. *Comparison of SVR kernel models with various split ratios*

Metrics	SVR Kernel Models	90:10	80:20	70:30	60:40	50:50
MAE	Wavelet	1.2067	3.859	16.808	37.911	56.676

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	EHT-BO	**0.13836**	0.3841	0.32521	1.4468	1.676
	CPR-GS	0.41886	**0.16337**	**0.18142**	0.15909	0.45432
	HAPR-GS	0.58358	0.50406	0.75457	**0.03729**	**0.01852**
MSE	Wavelet	2.3187	20.5	68.88	195.4	567.2
	EHT-BO	**0.19246**	0.25516	0.21887	3.7253	5.4118
	CPR-GS	0.24127	**0.05173**	**0.094347**	**0.097098**	**0.7146**
	HAPR-GS	0.54848	0.47942	1.1292	0.34318	0.73642
RMSE	Wavelet	1.5227	4.5277	19.206	46.855	75.281
	EHT-BO	**0.17101**	0.50513	0.46784	1.9301	2.3263
	CPR-GS	0.49119	**0.22744**	**0.30716**	**0.31161**	0.84534
	HAPR-GS	0.74059	0.6924	1.0627	0.58582	**0.27137**
MAPE	Wavelet	0.96371	2.842	15.513	42.663	77.395
	EHT-BO	**0.10268**	0.3098	0.3087	1.6119	2.1876
	CPR-GS	0.33149	**0.13438**	**0.14173**	**0.13115**	0.38811
	HAPR-GS	0.44751	0.41612	0.71648	0.43222	**0.25465**
R^2-Score	Wavelet	0.99699	0.97981	0.91588	0.84546	0.80359
	EHT-BO	**0.99996**	0.99975	0.99989	0.99865	0.99818
	CPR-GS	0.99969	**0.99978**	**0.99995**	**0.99996**	0.99976
	HAPR-GS	0.99929	0.99938	0.99944	0.99964	**0.99982**

A comparison was made between wavelet, EHT-BO, CPR-GS, and HAPR-GS using regression metrics (MAE, MSE, RMSE, MAPE, and R^2 score) to predict the closing price of Amazon stocks using various split ratios. Based on metrics the SVR models obtained results are shown in Table 1.

With a 90:10 split ratio, the EHT-BO method achieved the best MAE, MSE, RMSE, MAPE, and R^2 scores at 0.138, 0.192, 0.171, 0.102, and 0.9999, respectively. Similarly, HAPR-GS generated results of 0.018, 0.736, 0.271, 0.254, and 0.9998 for 50:50 split ratios. CPR-GS achieved better results for the remaining split ratios compared to other methods. Based on the CPR-GS method, the most promising values were 0.159, 0.051, 0.227, 0.131, and 0.9999 in the 80:20, 70:30, and 60:40 split ratios. The visual representation of these results is represented in Figures 5(a)–5(e).

4.2 Discussion

The results suggest that the EHT-BO method is well-suited for the 90:10 split ratio due to the data's non-linearity (absence of a straight-line relationship). Conversely, the HAPR-GS method might be appropriate for the 50:50 split, as the data likely contains both linear and non-linear relationships. For the 80:20, 70:30, and 60:40 splits, the CPR-GS method appears most effective due to the data's complex non-linearity (beyond simple curves or straight lines).

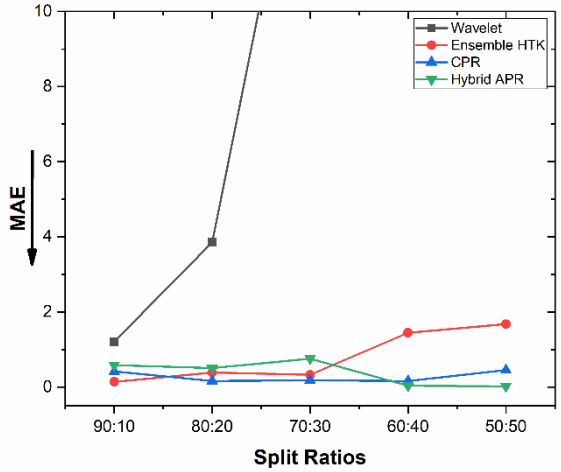


Figure 5(a). *MAE of four SVR kernel models*

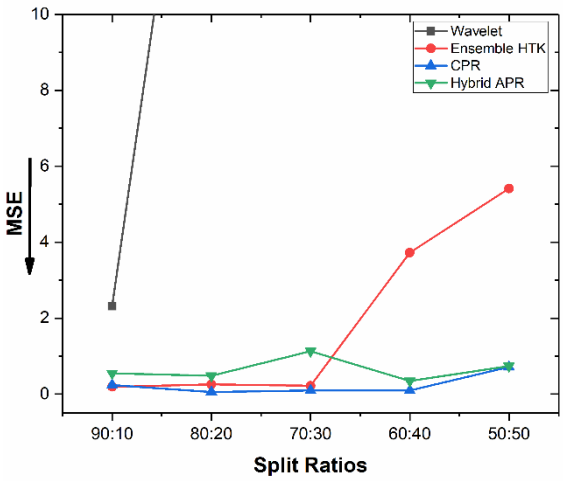


Figure 5(b). *MSE of four SVR kernel models*

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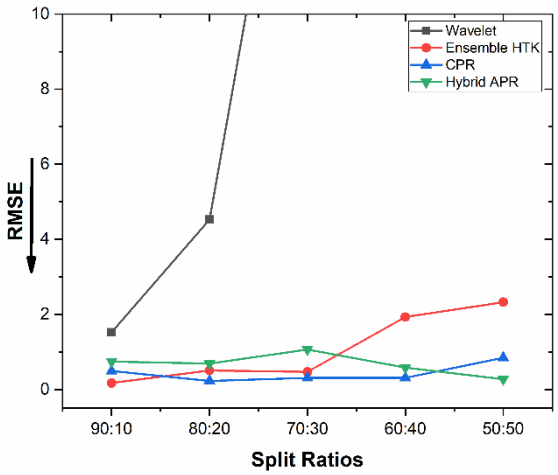


Figure 5(c). *RMSE of four SVR kernel models*

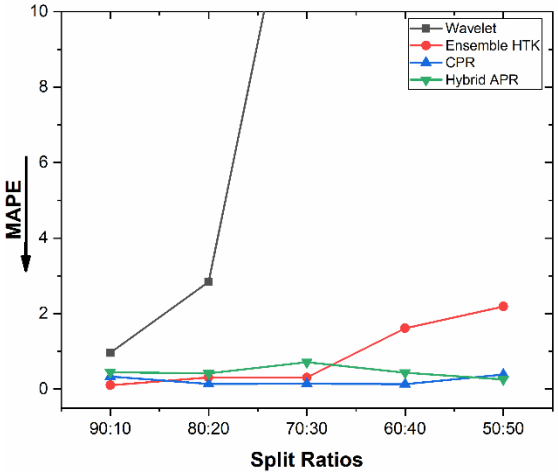


Figure 5(d). *MAPE of four SVR kernel models*

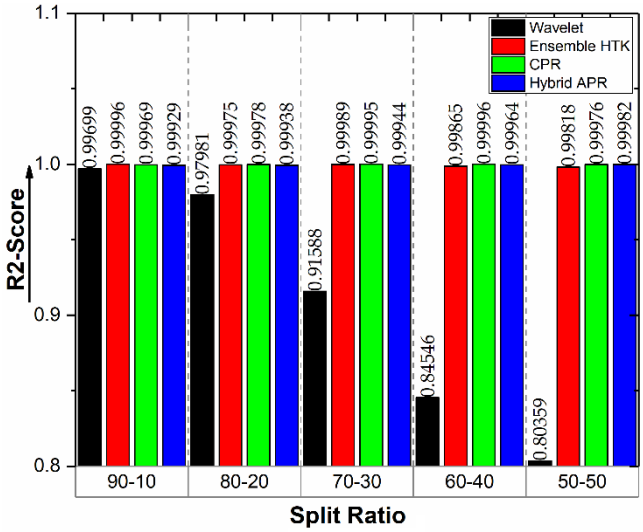


Figure 5(e). *R^2 score of four SVR kernel models*

Figure 6 illustrates these various data relationships. Notably, researchers often favor split ratios between 80:20 and 70:30 for optimal results, and the CPR-GS method achieved superior performance at these ratios compared to other kernel methods.

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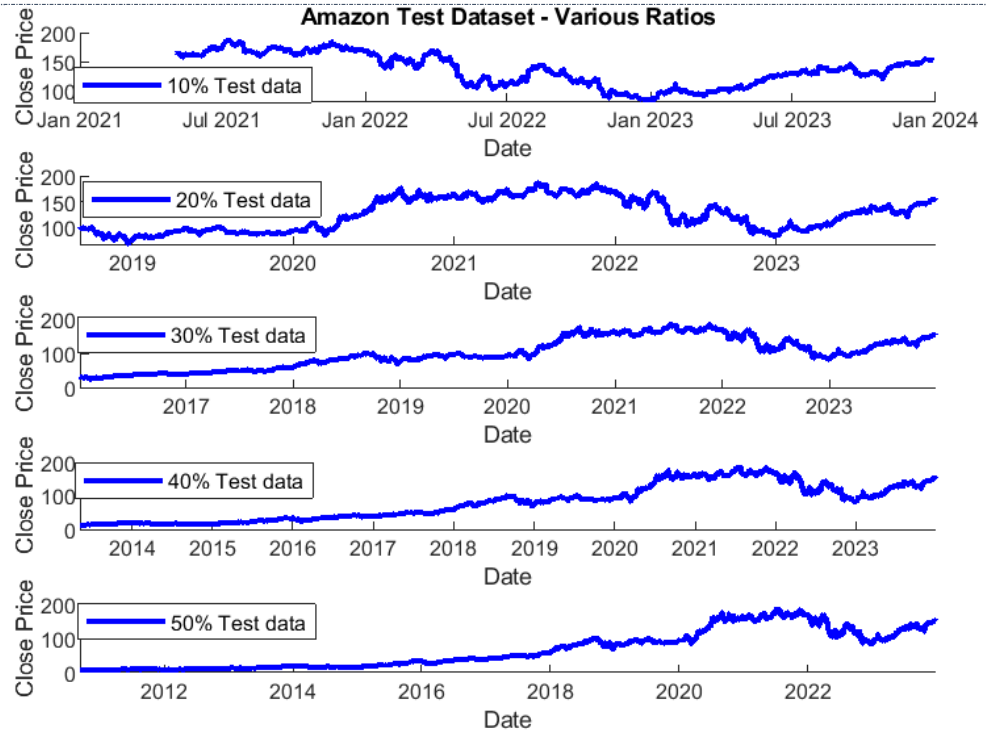


Figure 6. *Actual curves for various split ratios of Amazon test dataset*

5. CONCLUSION

This paper compares new kernel methods for Support Vector Regression (SVR) using Amazon's daily historical data to predict stock prices. An SVR algorithm was implemented with four kernel models: wavelet, EHT-BO, CPR-GS, and HAPR-GS. The proposed kernel models were evaluated using regression metrics: Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and Coefficient of Correlation (R^2 score). Among the SVR models with different kernels, the CPR-GS method achieved the best performance, with MAE of 0.159, MSE of 0.052, RMSE of 0.227, MAPE of 0.131, and R^2 score of 0.999. Future research can explore alternative kernels, combinations of models, and incorporate more diverse features and data sources for time series data prediction.

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